

COMP 141: Introduction to Computing

Filesystem Playground: Student Reference Guide

Directory Hierarchy & File Reference for Bob's and Alice's Workspaces

1 About Bob and His Files

Welcome to the **Filesystem Playground**! This playground simulates the personal Unix home directory of an organized undergraduate computing student named **Bob**. You can use this environment to explore, inspect, and practice working with directory hierarchies, navigation commands, file viewers, and text-processing tools from the Linux shell.

1.1 Who is Bob?

Bob is a sophomore student taking introductory courses in computing. He is methodical and keeps his home directory neatly structured into logical directories and subdirectories:

- **Academics** (`~/academics/`): Bob keeps his coursework organized by class. He is currently enrolled in **CS 141** (Linux Shell & CLI) and **CS 142** (Introduction to Python Programming). For each class, Bob maintains lecture notes, homework exercises, and lab workspaces.
- **Personal Projects** (`~/projects/`): In his free time, Bob writes practical automation tools in Bash (`shell_tools/`) and text-based terminal games in Python (`python_games/`). None of his scripts are dangerous or complicated; they are straightforward scripts just a step or two above “Hello World.”
- **Personal Hobbies** (`~/hobbies/`): Outside of computing, Bob is an avid distance runner and poetry enthusiast:
 - **Running**: Bob trains for distances ranging from 10K races to half-marathons. He logs his daily mileage in a master CSV spreadsheet, tracks running shoe mileage, follows a 12-week half-marathon plan, and saves race logistics and personal records.
 - **Poetry**: Bob studies the works of poets like Robert Frost, Emily Dickinson, and Langston Hughes. He also writes original poetry and keeps working drafts of verse in various stages of completion.
- **Environment Configuration** (`~/.ssh/`, **dotfiles**): Bob uses a standard Bash shell and OpenSSH for remote server connections. He maintains standard shell startup files (`.bashrc`, `.bash_profile`, `.bash_logout`) and SSH credentials.

1.2 Who is Alice?

Alice is Bob's classmate and fellow sophomore computing student. Like Bob, she takes introductory computing classes, but her workflow and interests reflect distinct personality traits and habits:

- **Academic Track**: Alice is enrolled in **CS 141** (Linux Shell), **CS 142** (Python Programming), and **Math 201** (Discrete Mathematics). She keeps structured notes, lab journals, and practice scripts for each course.
- **Compact Hierarchy Structure**: Unlike Bob's 4-level deep hierarchy, Alice prefers a shallower hierarchy of **maximum depth 3**, with at most 3 subdirectories per folder.
- **Purposeful Hidden Files**: Alice frequently uses hidden dotfiles (`.*`) not just for shell configs, but also for confidential personal notes, exam review sheets, and scratch files.
- **Personal Hobbies (Tabletop RPGs & Modular Synthesizers)**:
 - **Electronic Music**: Alice produces electronic music and designs analog synthesizer patches. She maintains a dedicated directory (`~/hobbies/music/`) for oscillator settings, audio frequency reference charts, and Python MIDI generators.
 - **Tabletop Gaming**: Alice is an avid Game Master for tabletop role-playing games (such as D&D) and collects modern board games. She tracks campaign storylines, character sheets, and Monte Carlo probability simulations.
- **File Varieties & Links**: Alice has no symbolic or hard links in her hierarchy. Her files are roughly 80% human-readable, with a few larger reference files and media assets.

1.3 Alice & Bob's Collaboration and Shared Files

In modern multi-user computing environments, students frequently collaborate on team projects while maintaining private individual accounts:

- **Shared Group (aliceandbob):** Alice and Bob are both members of a shared supplementary group named `aliceandbob`. The student account running the playground is also enrolled in this group.
- **Shared Folder (~/.alice/shared/):** Alice hosts a shared collaborative directory inside her workspace. The directory is owned by `alice:aliceandbob` with `setgid` permissions (2775), allowing both Alice and Bob to read, write, and execute files within it.
- **Cross-User Ownership:** The shared directory contains files authored by Alice (`alice:aliceandbob`), files contributed by Bob (`bob:aliceandbob`), and unshared private drafts (`alice:alice` and `bob:bob`).
- **Cross-Tree Sharing:** Bob also shares select files from his own hierarchy (such as course cheatsheets and original poetry) with group `aliceandbob`.

Architectural Metric	Bob's Hierarchy (~/.BobsFiles)	Alice's Hierarchy (~/.alice/)
Maximum Tree Depth	Depth 4	Depth 3
Subdirectories per Level	Up to 4 children per node	At most 3 children per node
Directory Count	25 directories	10 directories
File Count	114 files	54 files
Links	2 symbolic links, no hard links	No links of any kind
Hidden-to-Visible Ratio	Mostly visible, a few hidden	Slightly favors visible (higher hidden ratio)
Primary Hobbies	Distance Running, Classic Poetry	Electronic Music, Tabletop Gaming
Multi-User Collaboration	Shares files via group <code>aliceandbob</code>	Hosts shared folder (<code>shared/</code> , 10 files)

1.4 Types of Files in the Hierarchy

As you navigate Bob's files, you will encounter a variety of file formats and conventions:

- **Markdown Notes (.md):** Bob writes extensive course notes and study guides in Markdown. These are plain text files that can be viewed with tools like `cat`, `less`, or `head`.
- **Redirected Command Outputs (.txt, .log):** Most of Bob's plain text files were created by redirecting output from Linux shell commands or small helper scripts (such as `df -h > disk_usage.txt` or `uptime > uptime_log.txt`).
- **Scripts (.sh, .py):** Executable shell and Python programs for routine tasks and introductory assignments.
- **Structured Data (.csv):** Comma-separated data tables used for running logs, grade tracking, and game scores.
- **Symbolic Links (current_training.csv, latest_lecture.md):** Pointers linking to active files located elsewhere in the tree.
- **Hidden Files (.*):** Files whose names begin with a period, including shell configurations and private notes, visible when using the `-a` flag with `ls`.
- **Binary & Media Files (.pdf, .png, .jpg, .tar.gz, .db):** Syllabi, course archives, route maps, and photo assets that cannot be read as plain text.

1.5 Setting Up the Playground (build_the_bob.sh)

To practice navigating and manipulating Bob's files, you can generate his complete directory structure directly inside your personal user account using the automated setup script `build_the_bob.sh`. The script creates a dedicated, self-contained root directory named `~/BobsFiles` containing all 25 directories and 114 files without modifying any of your own files.

Step 1: Download the Script with `curl`

The script is hosted publicly on Google Drive (Public File ID: 11qwVImYoeAhZRQYS8Jrtgj5GcEomCAN8) and can be retrieved using the shortened URL <https://tinyurl.com/23vtqkux>. Use `curl` to fetch the script from the web. The `-L` flag instructs `curl` to follow HTTP redirects, while `-o build_the_bob.sh` specifies the destination file name:

```
curl -sSL -o build_the_bob.sh "https://tinyurl.com/23vtqkux"
```

Command breakdown: `-s` enables silent mode (suppressing progress meters), `-S` ensures error messages are still displayed if something fails, `-L` follows redirects, and `-o build_the_bob.sh` saves the output into your current directory.

Step 2: Make the Script Executable with `chmod`

Files downloaded from the web typically arrive with read-and-write permissions only (`rw-r--r--` or numeric mode `644`). Before the shell can execute the script, you must add execute (`x`) permission for yourself (the file owner / user, `u`) using `chmod`:

```
chmod u+x build_the_bob.sh
```

You can verify the file permissions updated to `-rwxr-xr-x` by running:

```
ls -l build_the_bob.sh
```

Step 3: Run and Manage the Playground

Execute the script from your terminal using its relative path (`./build_the_bob.sh`). The script supports both **Single-User Mode** (for Problem Sets 1 & 2) and **Multi-User Mode** (for Problem Set 3):

```
# -- Single-User Mode (Bob's Files Only) --
./build_the_bob.sh          # Builds ~/BobsFiles (default, no sudo needed)
./build_the_bob.sh -r      # Resets ~/BobsFiles (prompts confirmation)
./build_the_bob.sh -d      # Deletes ~/BobsFiles entirely

# -- Multi-User Mode (Bob & Alice Multi-User Hierarchy) --
sudo ./build_the_bob.sh -a  # Builds ~/alicebob_home with bob & alice accounts
sudo ./build_the_bob.sh -a -r # Resets ~/alicebob_home to clean state
sudo ./build_the_bob.sh -a -d # Deletes ~/alicebob_home and prompts to remove accounts
sudo ./build_the_bob.sh -cleanup-users # Removes bob, alice, and aliceandbob accounts
```

- **Single-User Playground (`~/BobsFiles`):** Generates Bob's complete 25 directories and 114 files directly within your normal user account. Requires no administrative privileges.
- **Multi-User Playground (`~/alicebob_home`):** Activated using `-a` or `-alice`. Provisions local system user accounts `bob` and `alice`, creates the shared group `aliceandbob`, and populates `alicebob_home/bob/` and `alicebob_home/alice/`.
- **UID/GID Persistence (`~/.bobenv`):** When multi-user mode is first run, assigned UIDs and GIDs are saved into a hidden file named `~/.bobenv`. If accounts are deleted and recreated, the exact same IDs are preserved.
- **Group Activation in Active Shells:** When your account is added to `aliceandbob`, Linux applies supplementary group tokens to new sessions. To activate the new group in an existing terminal without logging out, run:

```
newgrp aliceandbob
```

Step 4: Start Exploring

Once built, navigate into Bob's directory to begin your lab activities:

```
cd ~/BobsFiles  
ls -la
```

2 Visual Directory Hierarchy (Forest Trees)

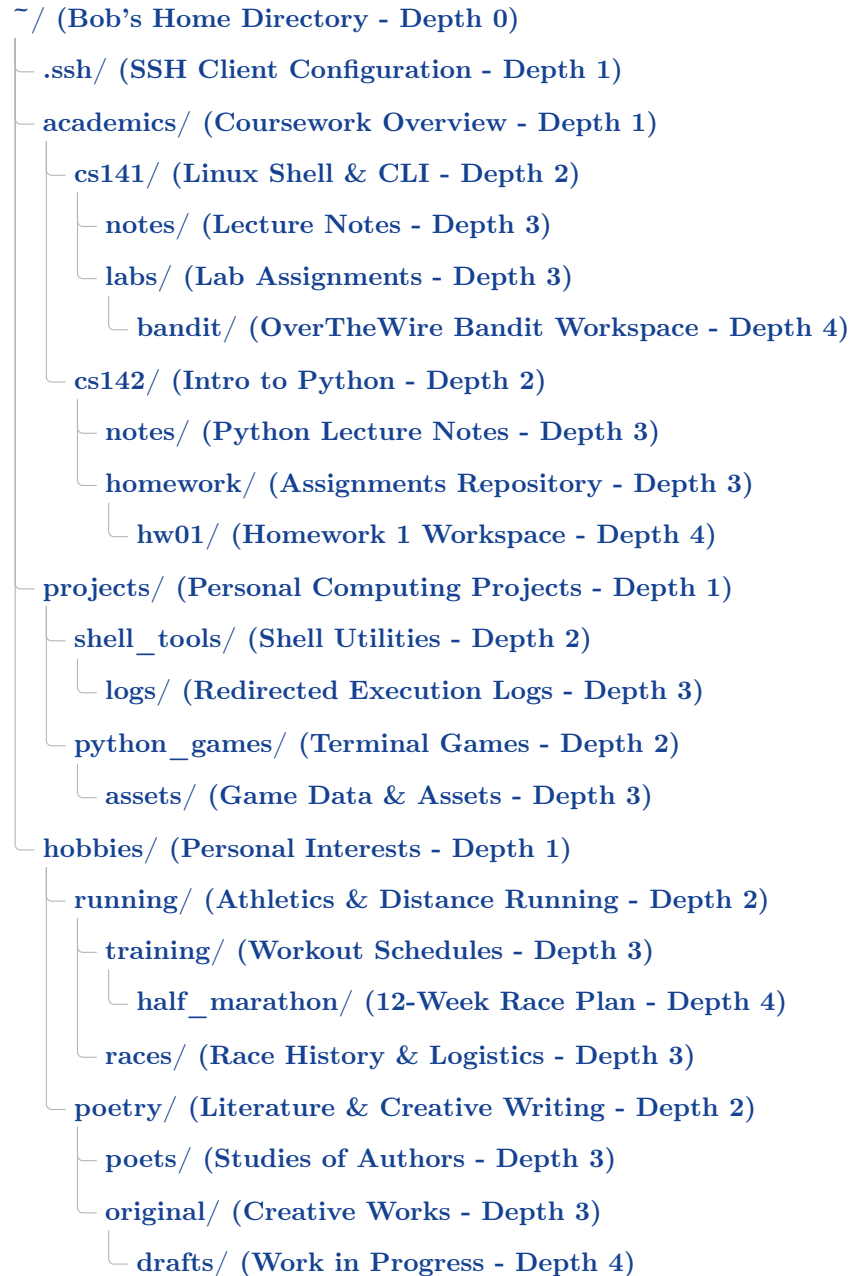
2.1 Visual Convention & Legend

In the directory forest trees below, nodes are color-coded to communicate their filesystem properties:

name/	Directory node	.name	Hidden file (.*)
name.ext	Markdown, plain text, CSV	name → target	Symbolic link
name.sh	Bash shell script	name.bin	Binary / archive / media / bytecode
name.py	Python script	name	Special character filename (spaces, symbols)

2.2 Master Directory Overview Tree

The complete directory topology across all 25 directories is illustrated below:



2.3 Branch A: Root Environment & SSH Configuration

This branch encompasses Bob's top-level user account configuration, personal overview files, and OpenSSH keys.

```
~/ (Depth 0)
├── .ssh/ (Depth 1)
│   ├── config      (SSH host configurations)
│   ├── id_ed25519   (Private authentication key; mode 600)
│   ├── id_ed25519.pub (Public Ed25519 key)
│   ├── authorized_keys (Authorized keys list)
│   └── known_hosts  (Host key fingerprints)
├── .bashrc          (Shell startup script with aliases)
├── .bash_profile     (Login script setting PATH)
├── .bash_logout      (Session cleanup)
├── .bash_history      (Recent command history)
├── README.txt        (Welcome guide to filesystem layout)
├── todo.md           (Weekly academic and personal task list)
└── current_training.csv → hobbies/running/training/half_marathon/weekly_schedule.csv
```

2.4 Branch B1: Academics — Linux Shell Course (CS 141)

The CS 141 branch houses lecture notes, lab materials, and Bandit wargame notes down to depth 4.

```
cs141/ (Depth 2)
├── notes/ (Depth 3)
│   ├── lecture01_intro_shell.md    (Shell syntax, options, args)
│   ├── lecture02_filesystem.md    (Paths, tree hierarchy, pwd, cd)
│   ├── lecture03_manipulation.md   (cp, mv, rm, wildcards)
│   ├── lecture04_redirection.md    (stdout, stdin, stderr, pipes)
│   ├── latest_lecture.md → lecture04_redirection.md
│   └── -notes_archive.txt         (Past topics outline)
├── labs/ (Depth 3)
│   ├── bandit/ (Depth 4)
│   │   ├── bandit00_log.txt        (Redirected bandit0 connection log)
│   │   ├── bandit01_notes.md       (Solution notes for level 1)
│   │   ├── bandit02_notes.md       (Solution notes for level 2)
│   │   ├── .bandit_passwords.txt   (Hidden password store)
│   │   └── level3_data.bin         (Dummy raw binary data challenge)
│   ├── lab_instructions.md         (Grading rubric and report instructions)
│   ├── ssh_test_output.txt         (Redirected output of ssh test)
│   ├── submission_template.txt     (Clean lab writeup template)
│   └── lab_materials.tar.gz        (Starter files archive)
├── cs141_syllabus.pdf              (Course syllabus document)
├── cheatsheet.md                   (Linux command quick reference)
├── grade_tracker.csv               (Lab and quiz grades table)
└── useful_aliases.sh               (Shell aliases for course tasks)
```

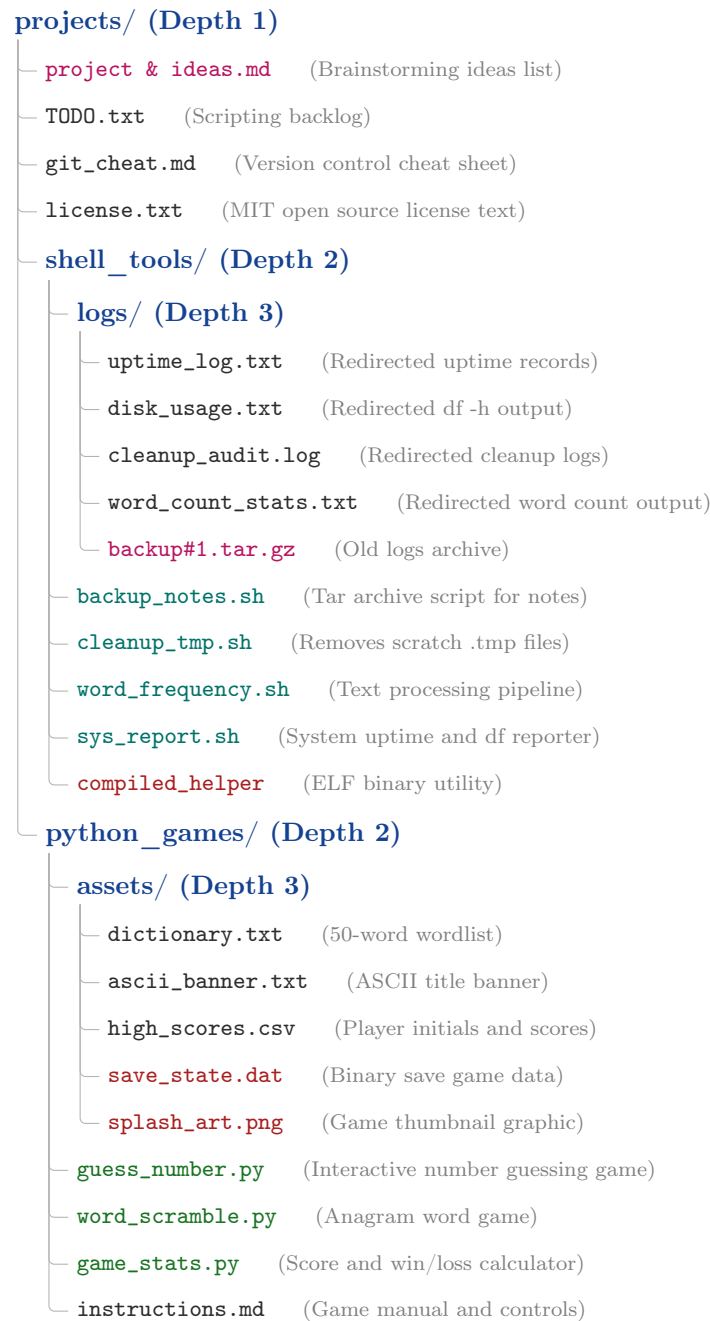
2.5 Branch B2: Academics — Intro to Python Course (CS 142)

The CS 142 branch houses Python lecture notes, exercises, and homework workspaces down to depth 4.

```
cs142/ (Depth 2)
├── notes/ (Depth 3)
│   ├── variables_and_types.md    (Data types, casting)
│   ├── conditionals.md          (if-elif-else logic)
│   ├── loops_and_iteration.md    (for, while, range)
│   ├── example_loop.py          (Accumulator pattern script)
│   └── control_flow.png          (Branching logic diagram)
├── homework/ (Depth 3)
│   ├── hw01/ (Depth 4)
│   │   ├── hw01_solution.py      (Temperature and tip calculations)
│   │   ├── test_hw01.py          (Unit assertions)
│   │   ├── test_results.txt       (Redirected test runner output)
│   │   ├── README.md             (Submission documentation)
│   │   └── hw01_bytecode.pyc      (Compiled bytecode fixture)
│   ├── assignment_guidelines.md  (PEP8 conventions)
│   ├── hw_schedule.txt           (Due dates and weights)
│   ├── rubric_template.md        (Peer review rubric)
│   └── .grading_secret.txt        (Hidden test hints)
├── cs142_syllabus.pdf            (Python course syllabus)
├── python_setup.md              (Virtualenv and interpreter guide)
├── course_outline.md            (Topic calendar)
└── py_quickstart.py             (Basic Python hello and test)
```

2.6 Branch C: Projects (Shell Tools & Terminal Games)

The projects branch contains Bob's personal programming projects in bash scripting and Python.



2.7 Branch D1: Hobbies — Distance Running

The running sub-branch encompasses training schedules, race logs, and performance metrics down to depth 4.

```
running/ (Depth 2)
├── training/ (Depth 3)
│   ├── half_marathon/ (Depth 4)
│   │   ├── weekly_schedule.csv    (12-week schedule plan)
│   │   ├── pacing_strategy.md     (Race split strategy)
│   │   ├── course_elevation.png   (Elevation profile image)
│   │   └── training_plan.pdf      (Official plan document)
│   ├── weekly_summary.txt         (Redirected mileage summary)
│   ├── interval_workouts.md       (Track repeats and tempo workouts)
│   ├── recovery_guide.md          (Nutrition, stretching)
│   └── .running_goals.txt         (Hidden PR target times)
├── races/ (Depth 3)
│   ├── upcoming_10k.md            (Logistics and bib info)
│   ├── race_results (2025).txt    (Official split results)
│   ├── pr_records.md              (Personal bests across distances)
│   ├── finisher_medal.jpg         (10K medal photo)
│   └── route_map.png              (Course route diagram)
├── running_log.csv                (Master CSV of daily runs)
├── shoe_mileage.txt               (Redirected awk cumulative km)
├── heart_rate_zones.md            (Aerobic/threshold zones)
├── quick_log.sh                   (Appends row to running_log.csv)
└── running_tracker.db             (SQLite database cache)
```

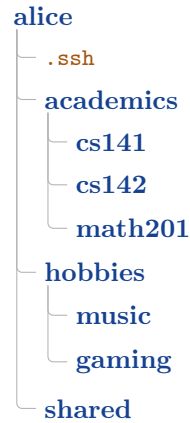
2.8 Branch D2: Hobbies — Poetry & Creative Writing

The poetry sub-branch models Bob’s creative writing, poet research, and drafts down to depth 4.



2.9 Alice's Master Directory Overview Tree

The directory tree below illustrates Alice's complete compact workspace (`~/alice/`), showing all 10 directories structured to a maximum depth of 3:



2.10 Alice Branch A: Root Environment & SSH Configuration



2.11 Alice Branch B: Academics (CS 141, CS 142 & Math 201)

```
academics
├── semester_schedule.txt
├── study_habits.md
├── .gpa_goals.md
├── .advisor_notes.txt
├── cs141
│   ├── lab_journal.md
│   ├── bandit_progress.txt
│   ├── shell_tricks.sh
│   ├── .exam1_review.md
│   └── .grading_criteria.txt
├── cs142
│   ├── recursion_notes.md
│   ├── calc.py
│   ├── quiz_prep.py
│   ├── .solutions_stash.py
│   └── unit_tests.log
└── math201
    ├── truth_tables.md
    ├── proof_techniques.md
    ├── logic_exercises.txt
    ├── .hw_answers.bak
    └── relations_diagram.png
```

2.12 Alice Branch C: Hobbies (Electronic Music & Tabletop Gaming)

```
hobbies
├── hobby_overview.md
├── weekend_plans.txt
├── .budget_allocations.csv
├── .reading_queue.txt
├── music
│   ├── synth_patch #1.txt
│   ├── frequency_guide.md
│   ├── ambient_track.py
│   ├── drum_loop.mp3
│   ├── synth_manual.txt
│   ├── .midi_mappings.txt
│   └── .favorite_artists.md
└── gaming
    ├── campaign notes (2026).md
    ├── d&d_char_sheet.txt
    ├── dice_odds_sim.py
    ├── .dm_secrets.txt
    └── boardgame_collection.csv
```

2.13 Alice Branch D: Shared Collaboration Space (~/.alice/shared/)

```
shared
├── project_proposal.md
├── joint_lab_report.md
├── study_guide_unit2.md
├── bandit_level105_hints.txt
├── synth_demo_stems.txt
├── bob_poetry_collab.md
├── game_rules_summary.txt
├── running_route_for_alice.csv
├── alice_private_draft.txt
└── bob_personal_memo.txt
```

3 Directory & File Catalog

This section provides an exhaustive inventory of all 25 directories and 114 files in Bob's home directory.

3.1 Root & Environment Configuration

1. Directory: ~/ (Home Root — Depth 0)

Subdirectories (4): .ssh/, academics/, projects/, hobbies/.

File Name	Type	Size	Description & Contents
.bashrc	Hidden Config	42 lines	Standard Bash user startup configuration file with aliases and prompt settings.
.bash_profile	Hidden Config	14 lines	Login shell startup script exporting environment variables (PATH, EDITOR).
.bash_logout	Hidden Config	6 lines	Console cleanup commands executed upon shell session exit.
.bash_history	Hidden Text	35 lines	Recorded command execution history from recent interactive terminal sessions.
README.txt	Plain Text	18 lines	Bob's orientation note outlining where his coursework and hobby files are kept.
todo.md	Markdown	24 lines	Weekly student task checklist with upcoming academic deadlines and workouts.
current_training.csv	Symbolic Link	1 link	Symbolic link pointing to <code>hobbies/running/training/half_marathon/weekly_schedule.csv</code> .

2. Directory: ~/.ssh/ (SSH Client Configuration — Depth 1)

Subdirectories (0): None (Leaf Directory).

File Name	Type	Size	Description & Contents
config	Plain Text / Config	16 lines	OpenSSH client configuration defining host aliases (cs-server, bandit) and port settings.
id_ed25519	Private Key	15 lines	OpenSSH Ed25519 private authentication key (standard mode 600 permissions).
id_ed25519.pub	Public Key	1 line	OpenSSH public key corresponding to Bob's private key.
authorized_keys	Plain Text	2 lines	Public keys authorized for inbound SSH connections to this account.
known_hosts	Plain Text	8 lines	Recorded cryptographic host fingerprints of trusted remote servers.

3.2 Academics: Coursework Branch

3. Directory: ~/academics/ (Academics Root — Depth 1)

Subdirectories (2): cs141/, cs142/.

File Name	Type	Size	Description & Contents
fall2026_schedule.txt	Plain Text	22 lines	Tabular listing of Fall 2026 courses, meeting times, building locations, and professors.
academic_goals.md	Markdown	20 lines	Bob's academic targets and personal learning objectives for the semester.
advising_notes.md	Markdown	25 lines	Meeting notes from advising appointments regarding major requirements and course electives.
calendar_export.ics	iCalendar	32 lines	Standard iCalendar format feed of academic deadlines exported from campus portal.

4. Directory: ~/academics/cs141/ (Linux Shell Course — Depth 2)

Subdirectories (2): notes/, labs/.

File Name	Type	Size	Description & Contents
cs141_syllabus.pdf	Binary (PDF)	48 KB	Official course syllabus document for CS 141.
cheatsheet.md	Markdown	32 lines	Quick reference sheet for frequently used Linux utilities and option flags.
grade_tracker.csv	CSV Text	16 lines	Comma-separated record of lab scores, reading question results, and quiz grades.
useful_aliases.sh	Shell Script	20 lines	Shell script defining custom command line aliases for course activities.

5. Directory: ~/academics/cs141/notes/ (Shell Lecture Notes — Depth 3)*Subdirectories (0):* None (Leaf Directory).

File Name	Type	Size	Description & Contents
lecture01_intro_shell.md	Markdown	26 lines	Course notes on terminal basics, shell syntax, command options, and arguments.
lecture02_filesystem.md	Markdown	28 lines	Course notes covering tree structure, absolute vs. relative paths, and navigation.
lecture03_manipulation.md	Markdown	30 lines	Notes on copying, moving, removing files, and wildcard pattern matching.
lecture04_redirection.md	Markdown	25 lines	Notes on standard streams (stdin, stdout, stderr), redirection operators, and pipelines.
latest_lecture.md	Symbolic Link	1 link	Relative symbolic link pointing directly to <code>lecture04_redirection.md</code> .
-notes_archive.txt	Plain Text	22 lines	Redirected summary of past lecture topics; filename begins with a hyphen.

6. Directory: ~/academics/cs141/labs/ (Shell Labs — Depth 3)*Subdirectories (1):* bandit/.

File Name	Type	Size	Description & Contents
lab_instructions.md	Markdown	24 lines	Lab report guidelines, submission checklist, and grading expectations.
ssh_test_output.txt	Plain Text	18 lines	Redirected output of SSH client version tests and server connectivity checks.
submission_template.txt	Plain Text	22 lines	Blank template for documenting lab commands and solution explanations.
lab_materials.tar.gz	Binary Archive	14 KB	Downloaded compressed tarball of starter lab materials from the course portal.

7. Directory: ~/academics/cs141/labs/bandit/ (Bandit Workspace — Depth 4)*Subdirectories (0):* None (Leaf Directory).

File Name	Type	Size	Description & Contents
bandit00_log.txt	Plain Text	16 lines	Redirected session recording from connecting to Bandit Level 0.
bandit01_notes.md	Markdown	20 lines	Bob's writeup on how to handle files named with leading dashes.
bandit02_notes.md	Markdown	22 lines	Bob's notes on quoting and escaping filenames that contain spaces.
.bandit_passwords.txt	Hidden Plain Text	12 lines	Hidden text file recording retrieved Bandit passwords for completed levels.
level3_data.bin	Binary Data	1 KB	Raw binary file simulating Bandit Level 3 challenge data.

8. Directory: ~/academics/cs142/ (Python Course — Depth 2)*Subdirectories (2):* notes/, homework/.

File Name	Type	Size	Description & Contents
cs142_syllabus.pdf	Binary (PDF)	52 KB	Official syllabus document for CS 142 (Introduction to Python).
python_setup.md	Markdown	24 lines	Installation guide for Python virtual environments and interpreter configuration.
course_outline.md	Markdown	22 lines	Weekly roadmap of upcoming programming topics and textbook readings.
py_quickstart.py	Python Script	18 lines	Simple Python test script checking library imports and basic math operations.

9. Directory: ~/academics/cs142/notes/ (Python Notes — Depth 3)*Subdirectories (0):* None (Leaf Directory).

File Name	Type	Size	Description & Contents
variables_and_types.md	Markdown	26 lines	Notes covering integers, floats, strings, booleans, and type conversion.
conditionals.md	Markdown	28 lines	Notes on comparison operators, boolean logic, and <code>if-elif-else</code> branching.
loops_and_iteration.md	Markdown	30 lines	Notes on <code>for</code> loops, <code>while</code> loops, and the <code>range()</code> built-in function.
example_loop.py	Python Script	16 lines	Demonstration script implementing a counter and accumulator loop pattern.
control_flow.png	Binary (Image)	28 KB	Flowchart diagram illustrating execution paths through conditional statements.

10. Directory: ~/academics/cs142/homework/ (Python Homework — Depth 3)*Subdirectories (1):* hw01/.

File Name	Type	Size	Description & Contents
assignment_guidelines.md	Markdown	22 lines	Instructor style guide outlining PEP 8 compliance, docstrings, and naming rules.
hw_schedule.txt	Plain Text	18 lines	Schedule of homework assignment release dates, due dates, and point weights.
rubric_template.md	Markdown	20 lines	Evaluation rubric detailing code quality, testing, and functionality criteria.
.grading_secret.txt	Hidden Plain Text	8 lines	Hidden text file containing sample test assertions from instructor office hours.

11. Directory: ~/academics/cs142/homework/hw01/ (Homework 1 — Depth 4)*Subdirectories (0):* None (Leaf Directory).

File Name	Type	Size	Description & Contents
hw01_solution.py	Python Script	32 lines	Bob's completed solution for Unit 1 arithmetic and temperature conversions.
test_hw01.py	Python Script	24 lines	Test script running assertions against Bob's solution functions.
test_results.txt	Plain Text	18 lines	Redirected test execution output from running <code>python3 test_hw01.py</code> .
README.md	Markdown	20 lines	Assignment writeup explaining program architecture and verification steps.
hw01_bytecode.pyc	Bytecode	900 bytes	Compiled Python bytecode generated during test execution.

3.3 Projects: Shell Tools & Terminal Games Branch**12. Directory: ~/projects/ (Projects Root — Depth 1)***Subdirectories (2):* shell_tools/, python_games/.

File Name	Type	Size	Description & Contents
project & ideas.md	Markdown	25 lines	Brainstorming list of utility scripts and text games Bob wants to build.
TODO.txt	Plain Text	18 lines	Actionable task backlog for personal coding improvements.
git_cheat.md	Markdown	22 lines	Quick reference guide on basic Git workflows (<code>status</code> , <code>add</code> , <code>commit</code>).
license.txt	Plain Text	21 lines	Standard MIT open-source license template applied to Bob's public repositories.

13. Directory: ~/projects/shell_tools/ (Shell Automation — Depth 2)*Subdirectories (1): logs/.*

File Name	Type	Size	Description & Contents
backup_notes.sh	Shell Script	22 lines	Utility script using <code>tar</code> to bundle lecture notes into an archive.
cleanup_tmp.sh	Shell Script	18 lines	Helper script using <code>find</code> and <code>rm</code> to purge scratch temporary files.
word_frequency.sh	Shell Script	16 lines	Text processing pipeline combining <code>tr</code> , <code>sort</code> , and <code>uniq -c</code> to count words.
sys_report.sh	Shell Script	20 lines	Diagnostic script logging date, host uptime, and free disk space.
compiled_helper	Executable Binary	16 KB	Compiled mock ELF 64-bit binary utility.

14. Directory: ~/projects/shell_tools/logs/ (Execution Logs — Depth 3)*Subdirectories (0): None (Leaf Directory).*

File Name	Type	Size	Description & Contents
uptime_log.txt	Plain Text	24 lines	Append log accumulating daily timestamps and uptime statistics.
disk_usage.txt	Plain Text	20 lines	Redirected snapshot of filesystem disk utilization produced by <code>df -h</code> .
cleanup_audit.log	Plain Text	18 lines	Redirected log capturing combined stdout and stderr from running <code>cleanup_tmp.sh</code> .
word_count_stats.txt	Plain Text	22 lines	Output file generated by <code>word_frequency.sh</code> on sample English text.
backup#1.tar.gz	Binary Archive	8 KB	Compressed archive of historical log files.

15. Directory: ~/projects/python_games/ (Terminal Games — Depth 2)*Subdirectories (1): assets/.*

File Name	Type	Size	Description & Contents
guess_number.py	Python Script	32 lines	Interactive CLI number guessing game with random number generator and loop.
word_scramble.py	Python Script	35 lines	Terminal anagram game reading candidate words from an external word list.
game_stats.py	Python Script	22 lines	Helper script calculating player win percentages and high scores from data files.
instructions.md	Markdown	20 lines	User manual explaining game commands, difficulty settings, and scoring rules.

16. Directory: ~/projects/python_games/assets/ (Game Assets — Depth 3)*Subdirectories (0): None (Leaf Directory).*

File Name	Type	Size	Description & Contents
dictionary.txt	Plain Text	50 lines	Plain text dictionary containing 50 sample words for <code>word_scramble.py</code> .
ascii_banner.txt	Plain Text	16 lines	ASCII art title graphic displayed when launching terminal games.
high_scores.csv	CSV Text	18 lines	Comma-delimited list of player initials, completion dates, and game scores.
save_state.dat	Binary Data	300 bytes	Serialized binary save file storing an in-progress game state.
splash_art.png	Binary (Image)	36 KB	Graphic thumbnail image designed for the project's documentation page.

3.4 Hobbies: Distance Running & Creative Poetry Branch

17. Directory: ~/hobbies/ (Hobbies Root — Depth 1)

Subdirectories (2): running/, poetry/.

File Name	Type	Size	Description & Contents
hobbies_overview.md	Markdown	22 lines	Bob's reflections on balancing computing coursework with athletics and creative arts.
gear_wishlist.txt	Plain Text	18 lines	Wishlist of running shoes, GPS watches, and recommended poetry books.
weekly_schedule.md	Markdown	24 lines	Time-blocking weekly schedule allocating time for classes, runs, and writing.
annual_budget.csv	CSV Text	16 lines	Annual budget tracking race registration fees, running gear, and book expenses.

18. Directory: ~/hobbies/running/ (Running & Athletics — Depth 2)

Subdirectories (2): training/, races/.

File Name	Type	Size	Description & Contents
running_log.csv	CSV Text	65 lines	Master log of daily training runs (date, distance in km, time, shoe model, heart rate).
shoe_mileage.txt	Plain Text	18 lines	Redirected output of an <code>awk</code> script calculating cumulative mileage on each shoe pair.
heart_rate_zones.md	Markdown	24 lines	Calculation of aerobic, tempo, and threshold training zones based on max heart rate.
quick_log.sh	Shell Script	20 lines	Interactive shell script prompting user for run parameters and appending to CSV.
running_tracker.db	SQLite DB	16 KB	SQLite 3 database file storing synced running activity metrics and splits.

19. Directory: ~/hobbies/running/training/ (Training Schedules — Depth 3)

Subdirectories (1): half_marathon/.

File Name	Type	Size	Description & Contents
weekly_summary.txt	Plain Text	24 lines	Weekly mileage totals and pace averages produced by Bob's analysis script.
interval_workouts.md	Markdown	22 lines	Descriptions of track interval sessions (800m repeats, mile repeats, tempo runs).
recovery_guide.md	Markdown	20 lines	Post-run recovery protocols covering hydration, stretching, and nutrition.
.running_goals.txt	Hidden Plain Text	12 lines	Hidden personal notes on target race times (10K: under 45 min; Half: under 1:45).

20. Directory: ~/hobbies/running/training/half_marathon/ (Race Plan — Depth 4)

Subdirectories (0): None (Leaf Directory).

File Name	Type	Size	Description & Contents
weekly_schedule.csv	CSV Text	28 lines	12-week half-marathon training progression (target of symlink <code>~/current_training.csv</code>).
pace_strategy.md	Markdown	24 lines	Mile-by-mile target pacing calculations and race fueling guidelines.
course_elevation.png	Binary (Image)	42 KB	Elevation profile graphic of the target half-marathon course.
training_plan.pdf	Binary (PDF)	64 KB	Official downloadable training plan document published by race organizers.

21. Directory: ~/hobbies/running/races/ (Race History & Events — Depth 3)*Subdirectories (0):* None (Leaf Directory).

File Name	Type	Size	Description & Contents
upcoming_10k.md	Markdown	22 lines	Logistics, bib pickup details, and wave start times for an upcoming 10K race.
race_results (2025).txt	Plain Text	26 lines	Official split times from Bob's previous competitive 10K event.
pr_records.md	Markdown	18 lines	Personal records across standard distances (5K, 10K, and 10 miles).
finisher_medal.jpg	Binary (Image)	52 KB	Photograph of Bob's finisher medal from the 2025 Autumn 10K race.
route_map.png	Binary (Image)	48 KB	Course map graphic showing turns, water stations, and elevation changes.

22. Directory: ~/hobbies/poetry/ (Poetry Studies & Writing — Depth 2)*Subdirectories (2):* poets/, original/.

File Name	Type	Size	Description & Contents
reading_list.md	Markdown	26 lines	Reading list of contemporary and classic poetry collections and anthologies.
poetic_forms.md	Markdown	28 lines	Reference notes on formal poetic structures (sonnets, haikus, villanelles, free verse).
word_associations.txt	Plain Text	24 lines	Brainstormed pairs of poetic imagery, evocative words, and metaphors.
anthology_cover.png	Binary (Image)	38 KB	Digital scan of the cover of Bob's favorite Robert Frost anthology.

23. Directory: ~/hobbies/poetry/poets/ (Poet Studies — Depth 3)*Subdirectories (0):* None (Leaf Directory).

File Name	Type	Size	Description & Contents
robert_frost.md	Markdown	28 lines	Biographical sketch and analysis of Frost's nature imagery in "The Road Not Taken."
emily_dickinson.md	Markdown	26 lines	Notes on Dickinson's concise meter, slant rhyme, and unconventional punctuation.
langston_hughes.md	Markdown	25 lines	Study of Hughes's role in the Harlem Renaissance and the rhythmic patterns of jazz poetry.
recitation.mp3	Binary (Audio)	96 KB	Short audio recording of a famous poem recitation.

24. Directory: ~/hobbies/poetry/original/ (Bob's Poetry — Depth 3)*Subdirectories (1):* drafts/.

File Name	Type	Size	Description & Contents
autumn_leaves.md	Markdown	22 lines	Finished short poem reflecting on the changing seasons across campus grounds.
terminal_solitude.md	Markdown	24 lines	Finished poem exploring late-night programming sessions and glowing screen light.
morning_run.md	Markdown	20 lines	Lyrical verse capturing the rhythm of running down mist-covered morning trails.
.secret_verse.txt	Hidden Plain Text	14 lines	Private, unfinished personal verse hidden from casual view.

25. Directory: ~/hobbies/poetry/original/drafts/ (In-Progress Poems — Depth 4)*Subdirectories (0):* None (Leaf Directory).

File Name	Type	Size	Description & Contents
My First Poem.txt	Plain Text	18 lines	Bob's earliest rhyming verse written during his first creative writing workshop.
haiku_ideas.md	Markdown	20 lines	Collection of experimental 5-7-5 syllable verses focusing on campus nature.
rhyme_scheme_test.txt	Plain Text	22 lines	Exploratory notes testing end-rhyme variations for a prospective sonnet.
abandoned_draft.bak	Backup Text	15 lines	Discarded stanza fragment saved with an editor backup extension.

3.5 Alice's Workspace Directory & File Catalog

The following tables document all 10 directories and 54 files within Alice's workspace, detailing file types, sizes, owner accounts, group attributions, and numeric permissions.

26. Directory: ~/alice/ (Alice's Root Directory — Depth 1)

Subdirectories (4): .ssh/, academics/, hobbies/, shared/.

File Name	Type		Owner:Group	Perms	Size	Description & Contents
.bashrc	Hidden Script	Shell	alice:alice	644	32 lines	Interactive shell aliases, magenta prompt, editor defaults.
.bash_profile	Hidden Script	Shell	alice:alice	644	10 lines	Login startup script that sources .bashrc and exports PATH.
.bash_logout	Hidden Script	Shell	alice:alice	644	5 lines	Clears console buffer on session termination.
.bash_history	Hidden Text	Plain	alice:alice	600	20 lines	Alice's terminal history across coursework, synth, and shared dirs.
welcome.txt	Plain Text		alice:alice	644	15 lines	Welcome message outlining major, courses, and workspace structure.
.todo_list.md	Hidden Mark-down		alice:alice	600	16 lines	Confidential weekly checklist for labs, gaming prep, and synth patches.

27. Directory: ~/alice/.ssh/ (SSH Credentials — Depth 2)

Subdirectories (0): None (Leaf Directory, Mode 700).

File Name	Type		Owner:Group	Perms	Size	Description & Contents
id_ed25519	Private Key		alice:alice	600	411 B	Alice's private Ed25519 authentication credential (restricted).
id_ed25519.pub	Public Key		alice:alice	644	100 B	Alice's public authentication key for campus servers.
known_hosts	Plain Text		alice:alice	644	2 lines	Host public keys for classroom Linux box and campus jump host.

28. Directory: ~/alice/academics/ (Academics Root — Depth 2)

Subdirectories (3): cs141/, cs142/, math201/.

File Name	Type		Owner:Group	Perms	Size	Description & Contents
semester_schedule.txt	Plain Text		alice:alice	644	18 lines	Weekly schedule of lectures, labs, music acoustics, and study group.
study_habits.md	Markdown		alice:alice	644	18 lines	Guidelines for focused study intervals, active recall, and terminal testing.
.gpa_goals.md	Hidden Mark-down		alice:alice	600	12 lines	Confidential GPA targets and STEM scholarship renewal thresholds.
.advisor_notes.txt	Hidden Plain Text		alice:alice	600	12 lines	Advising consultation notes on major requirements and electives.

29. Directory: ~/alice/academics/cs141/ (Linux Shell Course — Depth 3)*Subdirectories (0):* None (Leaf Directory).

File Name	Type	Owner:Group	Perms	Size	Description & Contents
lab_journal.md	Markdown	alice:aliceandbob	664	22 lines	Lab reflections on wildcards, streams, and Bandit levels (shared).
bandit_progress.txt	Plain Text	alice:alice	644	16 lines	Summary of completed OverTheWire levels and active Level 5 notes.
shell_tricks.sh	Executable Shell	alice:alice	755	28 lines	Script illustrating parameter defaults, file loops, and logging.
.exam1_review.md	Hidden Mark-down	alice:alice	600	20 lines	Study notes on octal modes, permission bits, and type vs which .
.grading_criteria.txt	Hidden Plain Text	alice:alice	644	10 lines	CS 141 syllabus grading breakdown across labs, exams, and projects.

30. Directory: ~/alice/academics/cs142/ (Python Course — Depth 3)*Subdirectories (0):* None (Leaf Directory).

File Name	Type	Owner:Group	Perms	Size	Description & Contents
recursion_notes.md	Markdown	alice:alice	644	20 lines	Conceptual guide to recursive base cases, call stacks, and tracing.
calc.py	Executable Python	alice:alice	755	35 lines	Command-line calculator supporting basic arithmetic and exponents.
quiz_prep.py	Executable Python	alice:alice	755	28 lines	Interactive self-test quizzer on Python builtins and data types.
.solutions_stash.py	Hidden Executable	alice:alice	755	24 lines	Alice's scratchpad solutions for recursion and string exercises.
unit_tests.log	Plain Text	alice:alice	644	10 lines	Captured console output from running pytest against homework 1.

31. Directory: ~/alice/academics/math201/ (Discrete Math — Depth 3)*Subdirectories (0):* None (Leaf Directory).

File Name	Type	Owner:Group	Perms	Size	Description & Contents
truth_tables.md	Markdown	alice:alice	644	22 lines	Truth tables for logical connectives, De Morgan's laws, equivalences.
proof_techniques.md	Markdown	alice:alice	644	20 lines	Templates for direct proof, contraposition, contradiction, induction.
logic_exercises.txt	Plain Text	alice:alice	644	18 lines	Practice problems analyzing binary relation properties and power sets.
.hw_answers.bak	Hidden Plain Text	alice:alice	600	10 lines	Backup scratch draft of parity proof for Homework 3.
relations_diagram.png	Binary (Image)	alice:alice	644	1x1 PNG	Reference diagram illustrating equivalence classes and partial orders.

32. Directory: ~/alice/hobbies/ (Hobbies Root — Depth 2)*Subdirectories (2):* music/, gaming/.

File Name	Type	Owner:Group	Perms	Size	Description & Contents
hobby_overview.md	Markdown	alice:alice	644	16 lines	Overview of sound design passions and tabletop RPG campaign leadership.
weekend_plans.txt	Plain Text	alice:alice	644	14 lines	Weekend schedule balancing studio tracking, gaming, and CS study.
.budget_allocations.csv	Hidden CSV Text	alice:alice	600	8 lines	Financial ledger of Eurorack modules, polyhedral dice, and board games.
.reading_queue.txt	Hidden Plain Text	alice:alice	600	6 lines	Reading list covering sound synthesis handbooks and game design.

33. Directory: ~/alice/hobbies/music/ (Electronic Music — Depth 3)*Subdirectories (0):* None (Leaf Directory).

File Name	Type	Owner:Group	Perms	Size	Description & Contents
synth_patch #1.txt	Plain Text	alice:alice	644	24 lines	Modular patch parameters: VCO mix, 24dB ladder filter, envelopes.
frequency_guide.md	Markdown	alice:aliceandbob	664	24 lines	Audio frequency reference chart (sub-bass to brilliance) shared with Bob.
ambient_track.py	Executable Python	alice:alice	755	30 lines	Script computing generative scale frequencies for ambient synthesizer drones.
drum_loop.mp3	Binary (Audio)	alice:alice	644	48 B	Audio sample asset header for percussive breakbeat rhythm.
synth_manual.txt	Plain Text	alice:alice	644	65 lines	Comprehensive reference guide on FM operator algorithms and index ratios.
.midi_mappings.txt	Hidden Plain Text	alice:alice	600	14 lines	Hardware CC knob assignments for cutoff, resonance, and envelope attack.
.favorite_artists.md	Hidden Markdown	alice:alice	600	10 lines	Historical synth pioneers and landmark albums that inspire Alice's music.

34. Directory: ~/alice/hobbies/gaming/ (Tabletop Gaming — Depth 3)*Subdirectories (0):* None (Leaf Directory).

File Name	Type	Owner:Group	Perms	Size	Description & Contents
campaign_notes (2026).md	Markdown	alice:alice	644	24 lines	Weekly campaign chronicle, adventuring party roster, and dungeon clues.
d&d_char_sheet.txt	Plain Text	alice:alice	644	26 lines	Character sheet for Kaelen Brightwood, High Elf Eldritch Knight Fighter.
dice_odds_sim.py	Executable Python	alice:alice	755	32 lines	Monte Carlo simulation tool for rolling dice with DC advantage/disadvantage.
.dm_secrets.txt	Hidden Plain Text	alice:alice	600	10 lines	Secret DM encounter keys, NPC betrayals, and dungeon trap locations.
boardgame_collection.csv	CSV Text	alice:alice	644	34 lines	Master spreadsheet cataloging 32 board games, complexity, and player counts.

35. Directory: ~/alice/shared/ (Shared Collaboration Space — Depth 2)*Subdirectories (0):* None (Leaf Directory, Mode 2775, Group aliceandbob).

File Name	Type	Owner:Group	Perms	Size	Description & Contents
project_proposal.md	Markdown	alice:aliceandbob	664	22 lines	Collaborative project proposal for joint command-line audio utility.
joint_lab_report.md	Markdown	bob:aliceandbob	664	22 lines	Co-authored report reviewing navigation tools, pipes, and security.
study_guide_unit2.md	Markdown	alice:aliceandbob	664	20 lines	Study outline for permissions, octal modes, and find queries.
bandit_level05_hints.txt	Plain Text	bob:aliceandbob	664	16 lines	Bob's search recommendations for finding ASCII files with exact byte size.
synth_demo_stems.txt	Plain Text	alice:aliceandbob	664	12 lines	Track notes and stem breakdown of synth layers shared with Bob.
bob_poetry_collab.md	Markdown	bob:aliceandbob	664	20 lines	Bob's technology poem ("Code & Silicon") shared for Alice's feedback.
game_rules_summary.txt	Plain Text	alice:aliceandbob	664	18 lines	Primer on d20 action economy and mechanics prepared for Bob.
running_route_for_alice.csv	CSV Text	bob:aliceandbob	664	8 lines	5K running loop across campus grounds mapped out by Bob.
alice_private_draft.txt	Plain Text	alice:alice	640	6 lines	Private reminder note left in shared folder (group read only).
bob_personal_memo.txt	Plain Text	bob:bob	644	6 lines	Bob's private workout reminder left in shared folder.